



CPSC 100

Computational Thinking

Sequential Algorithm + Programming

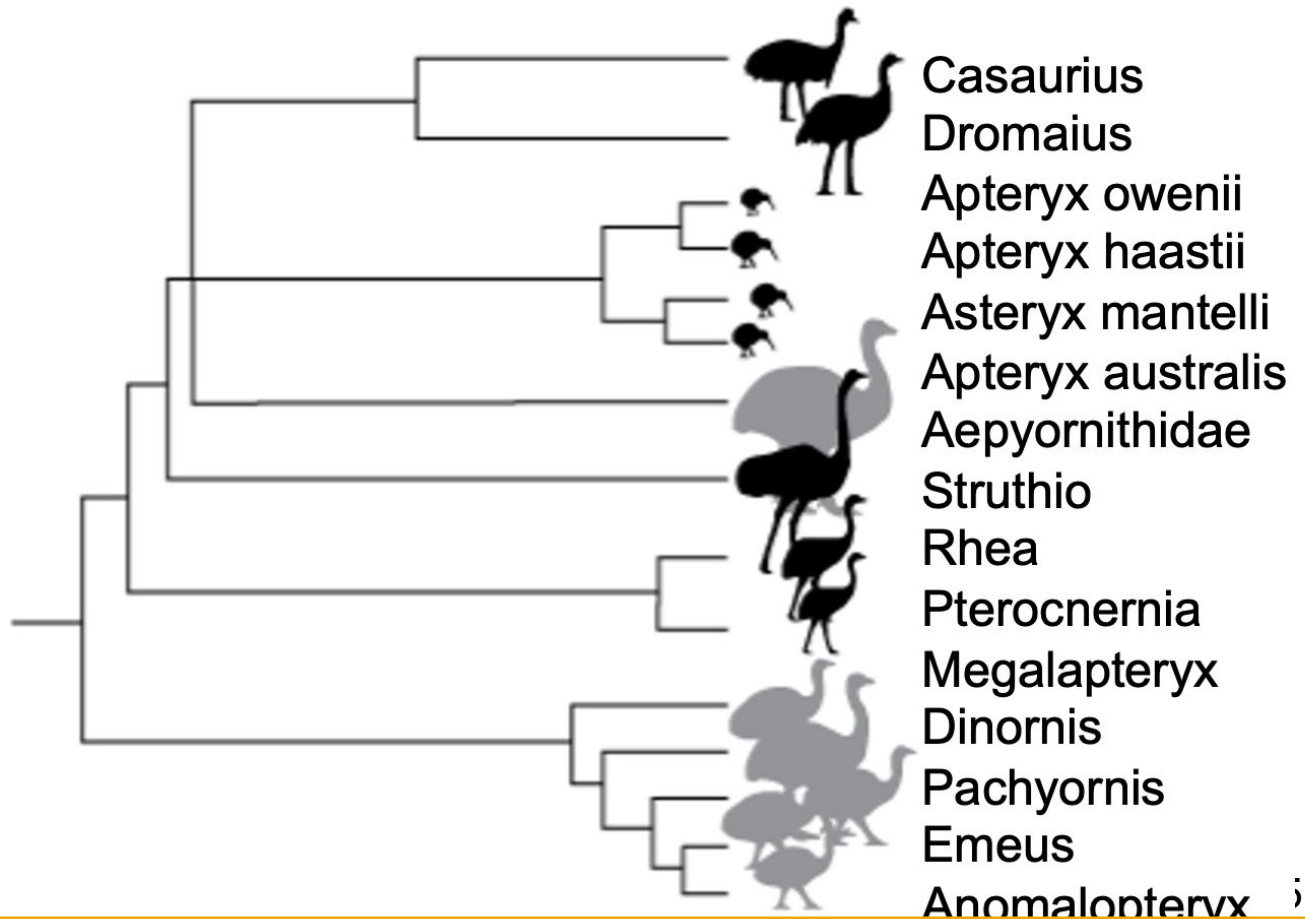
Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Slides for Pre-reading

**Slides with yellow borders
will not be covered in
class, but is still testable
content - you should
review this before class.**

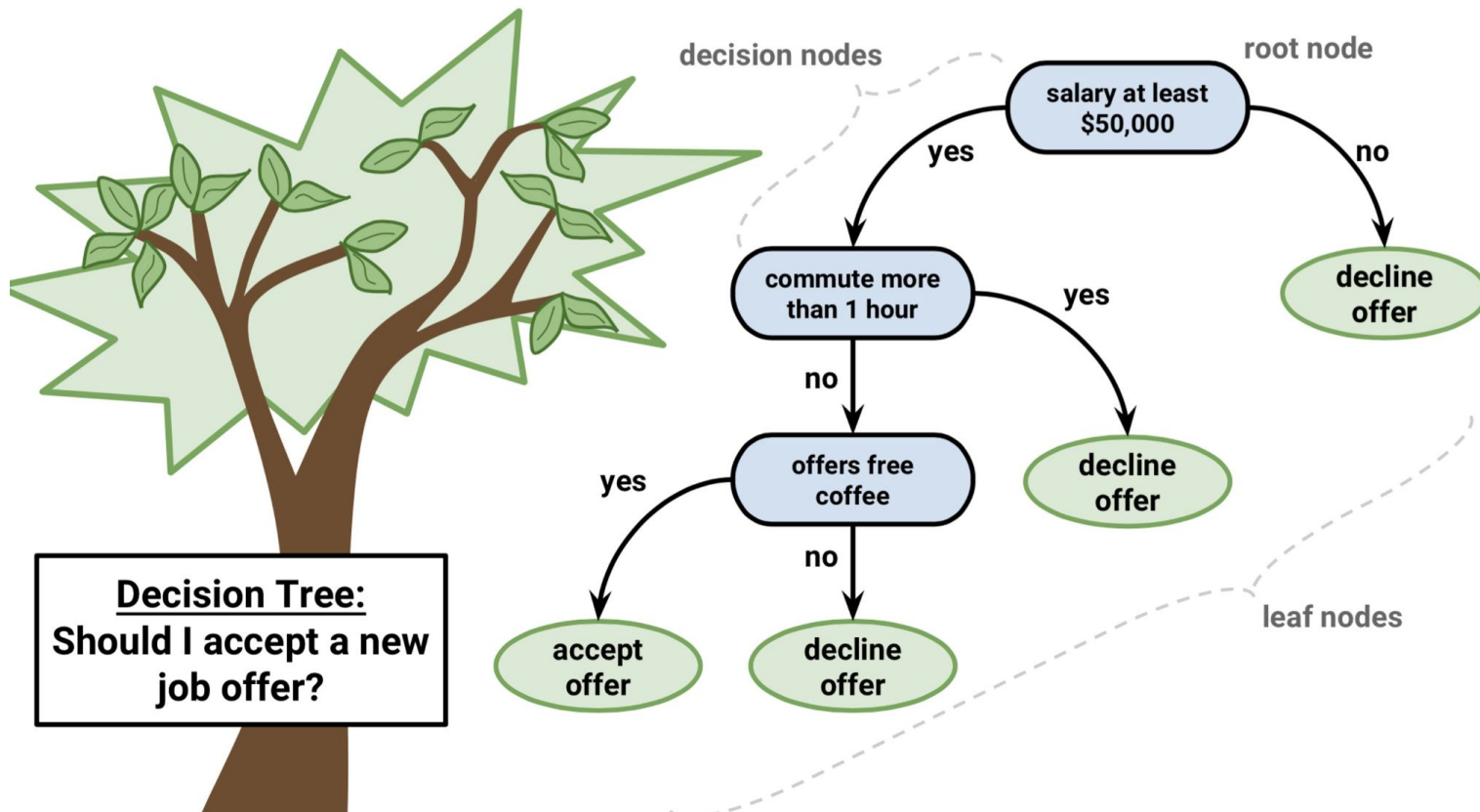
Non-CS Decisions Trees

**Rooted trees
in CS often
(but not always)
drawn with
root on top**



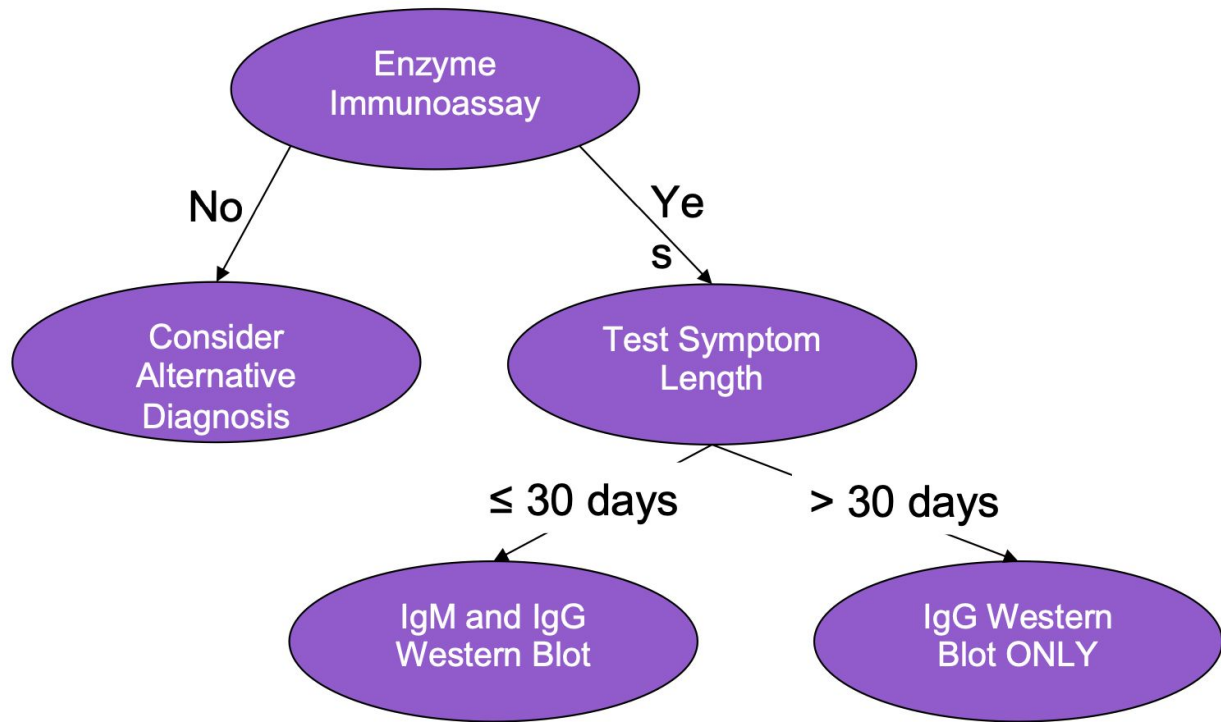
Decision trees

Trees whose node labels are **attributes**, edge labels are **conditions**



Decisions Trees in Medicine

Decision tree for
Lyme Disease
diagnosis



Decisions Trees in Business



Graziadio Business Review

A Peer-Reviewed Journal of Relevant Information and Analysis

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How Gerber Used a Decision Tree in Strategic Decision-Making

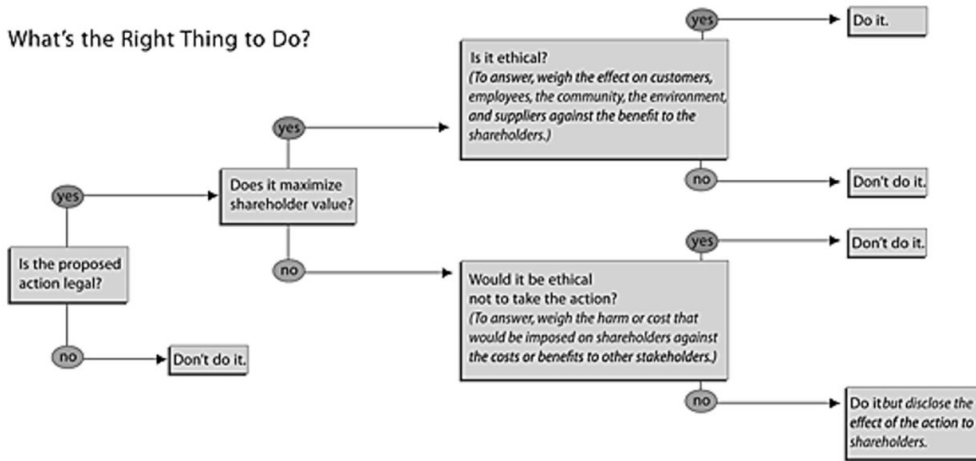
Possible outcomes explored in an in Products Safety Commission.

By JAY BUCKLEY and THOMAS J. DUDLEY, DBA

1999 Volume 2 Issue 3

Decision trees can assist executives in making strat

What's the Right Thing to Do?



Course Admin



Agenda

- Learning Goals
- Course Admin
- Sequential Algorithm
- Intro to Programming

Learning Goals



Learning Goals

After this week's lecture, you should be able to:

- Apply the sequential algorithm to spell a word and compute its cost.
- Apply the “Breaking Bad” algorithm to spell a word and compute its cost.
- Identify the differences between sequential and "Breaking Bad" algorithms
- Identify components of a block-based visual language (Snap!)

Clicker Questions (using Hands)



Clicker Question



Q: What is entropy in the context of decision trees?

- A. The uncertainty or mixed information in a dataset
- B. The number of nodes in a tree
- C. The difference between training and test data
- D. The length of the decision path in a tree



Clicker Question



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Clicker Question



Q: What is a classifier?

- A. Specific type of algorithm that makes image predictions
- B. An algorithm that maps input data to one or more categories
- C. A type of decision tree used for data mining
- D. A type of data storage for algorithms
- E. I don't know



Clicker Question



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A. Specific type of algorithm that makes image predictions

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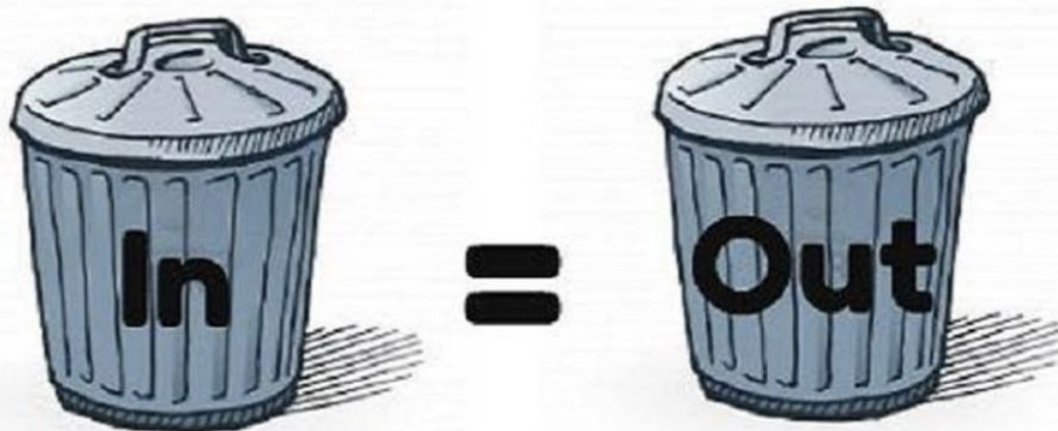
C. A type of decision tree used for data mining

D. A type of data storage for algorithms

E. I don't know

Data's influence on algorithms

Training data influences the classifier



Garbage in, Garbage out (GIGO)

Sequential Algorithm



Sequential Algorithm for Signaling Words

Jean-Dominique Bauby was a journalist who had *locked-in syndrome*.

He used the *partner-assisted scanning system* to dictate a 130 page book by blinking.





Sequential Algorithm for Signaling Words

“It is a simple enough system,” Bauby explains.

“You read off the alphabet... until, with a blink of my eye, I stop you at the letter to be noted. The maneuver is repeated for the letters that follow, so that fairly soon you have a whole word.”

In other words, you go through each letter, row by row, and read off all the letters.





Sequential Algorithm for Signaling Words

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0





Sequential Algorithm for Signaling Words

Let's set our goal to **minimize the number of letters that have to be looked** at to spell a word.

So each letter you look at has a cost of “1”

We also have to decide if there's any extra “cost” to having the “signals”.

Let's say each signal also costs “1”



Sequential Algorithm for Signaling Words

Sequential

8 to get to the letter F
+ 1 to signal "F"

9 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Q: What is the cost to signal the word “FAN”?



- A. 26
- B. 28
- C. 29
- D. 30
- E. 31

A	B	C	D	1	2
E	F	G	H	3	4
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Breaking Bad Algorithm





Example of cost counting: Letter F

2 to get to the “E” row (1)
1 to signal the “E” row (1)
2 to get to “F” in the row
+1 to signal “F”

6 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0



Breaking Bad Algorithm

Q: What is the cost to signal the word “FAN”?



iClicker

- A. 17
- B. 18
- C. 19
- D. 20
- E. 21

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Activity



Activity: Algorithms in Action

Find a word that works better the Sequential way.

Find a word that works better the Breaking Bad way.

Which algorithm is better and why?

Rules:

Both words must be at least 4 letters!

Use the same chart

Activity: Algorithms in Action

Breaking Bad vs. Sequential Algorithms

View... ▾

Breaking Bad vs. Sequential Algorithms

- Find a word that works better the Sequential way.
- Find a word that works better the Breaking Bad way.

Rules: Both words must be at least 4 letters! Use the same chart.

Overall, which algorithm is better and why?

ans.md



1

Discussion



Activity: Algorithms in Action

Your signal algorithms used

Decomposition: breaking the problem down into smaller tasks

Abstraction: describing the solution in a general way that's applicable no matter what order the letters/numbers are in the table



Extra Practice

Example of cost counting: Letter L

3 to get to the “I” row
 1 to signal the “I” row
 4 to get to “L” in the row
 +1 to signal “L”

9 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Example of cost counting: Letter O

4 to get to the “O” row
 1 to signal the “O” row
 1 to get to “O” in the row
 +1 to signal “O”

7 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Example of cost counting: Letter V

5 to get to the “U” row
 1 to signal the “U” row
 2 to get to “V” in the row
 +1 to signal “V”

9 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Example of cost counting: Letter E

2 to get to the "E" row
 1 to signal the "E" row
 1 to get to "E" in the row
 +1 to signal "E"

5 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0